

Expedient synthesis of β,β -disubstituted α -methylenepropionates

Kallolmay Biswas,^a Christoph Börner,^a Josepe Gimeno,^b Paul J. Goldsmith,^a
 Daniella Ramazzotti,^c Angela L. K. So^a and Simon Woodward^{a,*}

^a*School of Chemistry, Nottingham University, University Park, Nottingham NG7 2RD, UK*

^b*Departament de Química, Universitat Autònoma de Barcelona, Bellaterra, 08193 Barcelona, Spain*

^c*Dipartimento di Chimica, Università di Sassari, Via Vienna 2, 07100 Sassari, Italy*

Received 30 September 2004; revised 9 November 2004; accepted 2 December 2004

Available online 21 December 2004

Abstract—Baylis–Hillman alcohols are excellent sources of the allylic halides $\text{ArCH}=\text{CH}(\text{CH}_2\text{X})(\text{CO}_2\text{R})$ ($\text{X}=\text{Br}, \text{Cl}$; $\text{R}^1=\text{Me}, \text{Et}, \text{Bu}^t$). The *Z* double bond isomers are attained in high isomeric purity ($>14:1$, *Z/E*). The halides are chemo- and regiospecifically transformed into the acrylates $(\text{ArCHR}^2)\text{C}(\text{=CH}_2)(\text{CO}_2\text{R}^1)$ on treatment with $\text{Zn}(\text{R}^2)_2$ ($\text{R}^2=\text{Me}, \text{Et}, \text{CH}_2\text{TMS}, \text{CH}_2\text{SiMe}_2\text{OMe}$) or PrZnBr in the presence of catalytic amounts of copper(I) salts (0.5–20 mol%) in high yield.
 © 2004 Elsevier Ltd. All rights reserved.

1. Introduction

Acrylate fragments constitute exceptionally valuable building blocks in organic chemistry participating readily in many asymmetric transformations including Diels–Alder¹ and Michael addition² reactions. While simple acrylate and methylacrylate esters are commercially available surprisingly few good (whether racemic or stereoselective) routes exist to even the simplest disubstituted species **1**. The two limited literature approaches to **1** available are either to access it through γ -selective alkylations of organometallic allyl species (route ‘A’, $\text{M}=\text{MgX}, \text{ZnX}$)³ or by α -selective alkylations of organovinyl species (route ‘B’, $\text{M}=\text{ZnX}$).⁴ Approach ‘A’ often suffers from competing α -alkylation, leading to a mixture of regioisomers. The displacement of the secondary halide in route ‘B’ is limited to highly reactive allylic systems under palladium catalysis. We thought that umpolung versions of this chemistry based on organometallic additions of R^2M to allylic halides might offer a chemo- and regiospecific approach to **1**. Such a route could be attained conceptually through reliable conjugate addition and Baylis–Hillman chemistry (approach ‘C’). Preliminary studies by us had demonstrated the viability of a catalytic approach and also indicated that route C offers potential for asymmetric catalysis.⁵ However, the enantioselectivities attained in this early work were resistant to optimisation. This led us to identify optimal routes to the

racemates of **1** (especially in the cases where R^1 is an aryl group) and to identify conditions for the separation of their enantiomers to support later mass asymmetric ligand screening.

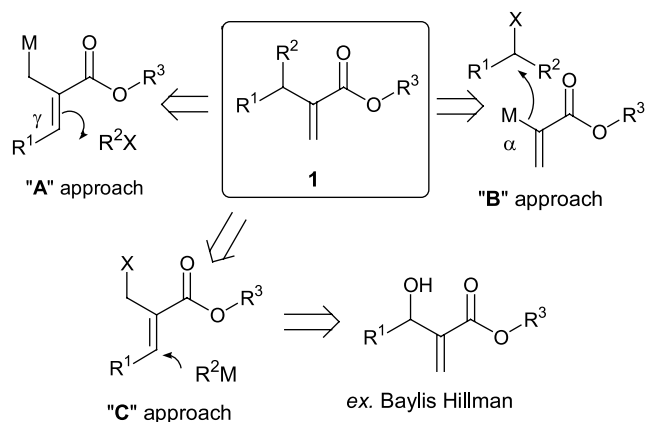
2. Results and discussion

2.1. Scope of the allylic halide ester

Due to the dearth of catalytic studies on Baylis–Hillman derived allylic halides (and pseudohalides) we wished to assure ourselves that the desired γ -specific catalytic addition was possible. As Kündig had demonstrated γ -specific addition of stoichiometric⁶ $\text{LiCuBu}_2\cdot\text{ZnCl}_2$ to related systems, we chose to concentrate on zinc and copper-based catalytic systems. A range of suitable Baylis–Hillman adducts **2** were prepared from ethyl acrylate and suitable aromatic aldehydes in the presence of stoichiometric DABCO. The slower reacting electron rich aldehydes could be co-promoted with lanthanum triflate.⁷ In all cases $>60\%$ conversion was attained in 7 days, the fastest reacting aldehyde (4- NO_2PhCHO) was essentially completely consumed within 8 h. Aromatic aldehydes were selected for convenience as Baylis–Hillman reaction of aliphatic aldehydes, while clean, is rather slow. Treatment of **2** with concentrated HBr or HCl , in the presence of H_2SO_4 , leads cleanly to either the bromides **3** or chlorides **4** (Scheme 1). A yield range of 60–95% is observed with the majority being over 90%. These eliminations proceed with very high *Z* selectivity ($>20:1$ for compounds **3**, $\sim 14:1$ for

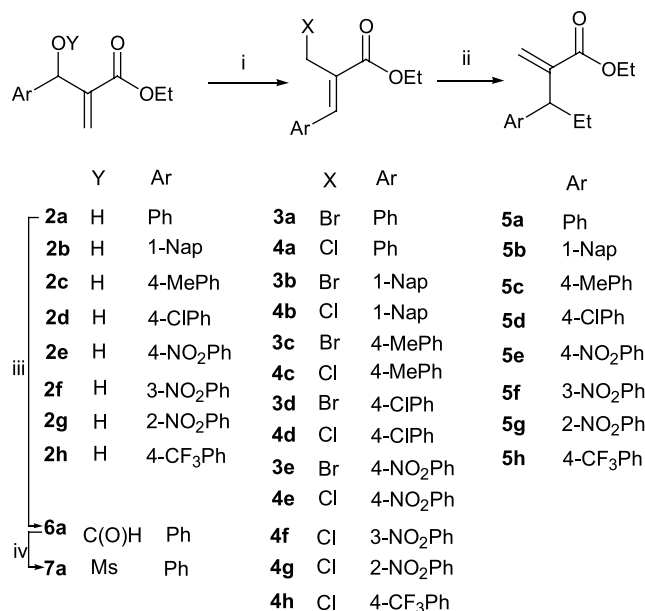
Keywords: Baylis–Hillman; $\text{S}_{\text{N}}2'$; Allylic substitution; Organozinc; Copper.

* Corresponding author. Tel.: +44 115 951 3541; fax: +44 115 951 3541;
 e-mail: simon.woodward@nottingham.ac.uk



Scheme 1.

4) originating from stereoelectronic controlled elimination of the zwitterionic intermediates.⁸ In many cases the crude halides **3–4** crystallise from Et₂O or Et₂O/pentane mixtures as geometrically pure species (Scheme 2).



Scheme 2. Reagents and conditions: (i) concd HX/H₂SO₄, 16–24 h, rt; (ii) ZnEt₂, THF, –20 °C, cat. [Cu(MeCN)₄]BF₄ (3 mol%) [for **3a–h**, **6a**, **7a**] or, cat. CuCN (20 mol%) and NBu₄Br (10 mol%) [for **4a–h**]; (iii) NEt₃/HCO₂H reflux; (iv) for **7a** treatment of **6a** with aq. HCl in methanol, followed by mesylation (MeSO₂Cl/NEt₃) of the derived alcohol.

Complete conversion of the bromides **3** to acrylates **5** could be achieved cleanly with 1.5 equiv of ZnEt₂ in the presence of [Cu(MeCN)₄]BF₄ (3 mol%). In general, the isolated yields were in the range 60–80%. The reaction of the bromides is very strongly promoted by [Cu(MeCN)₄]BF₄, and at 0.5 mol% loading complete conversion is attained. However, in the absence of copper(I) <15% conversions are seen in control reactions under otherwise identical conditions. The catalytic reactions of **3** are completely chemo- and regiospecific. The product propionates **5**, although powerful Michael acceptors in their own right, are not attacked under the reaction conditions. We speculate that bidentate coordination via both the ester function and

the halide is required by the catalyst. Alternatively, the enthalpy associated with formation of ZnCl₂ may provide the reason for the higher reactivity of **3/4**. No S_N2 products (resulting from attack of ZnEt₂ α to the leaving group) are apparent in the crude reaction products within the limits of ¹H NMR detection, behavior analogous to Kündig's stoichiometric system. One explanation for the reaction's regiospecificity may be that an addition–elimination mechanism akin to conjugate addition operates via an undetected enolate intermediate. Attempts to intercept such potential intermediates through addition of PhCHO to the reaction mixture were however unsuccessful.

The chlorides react much more slowly. For example, the reaction of **4e** is only 57% complete after 40 min at –20 °C. Use of an alternative catalyst system employing CuCN (20 mol%) and tetrabutylammonium bromide (TBAB) (10 mol%), allowed rapid conversions of the chlorides within 45 min at –20 °C. Several other quaternary ammonium salts also provide strong rate acceleration in these reactions. The reactions of **4** are completely clean – only **5** is formed and simply isolated by column chromatography. Additionally, it was found that use of P(OEt)₃ (4 mol%) strongly retards the rate of reaction (~50% conversion of **3a** after 40 min at 0 °C). Similarly, Feringa's phosphoramidate⁹ with Cu^I catalyst provided less than 3% conversion of **4** after 20 min at –20 °C, suggesting that chiral phosphorus species may not be useful in asymmetric variants of this reaction. Despite the wide difference in the electronic factors in substrates **3**, **4**, and **9**, all are effective coupling partners within each series, allyl halides bearing electron withdrawing substituents (–I, or –M) are the quickest reacting. In all the cases investigated no evidence was seen for attack at any of the aryl substituents in compounds **3** or **4**. For comparison of leaving group effects the formate **6a** was prepared by adaptation of known procedures, and from this the reactive mesylate **7a** also prepared. The former was unreactive under [Cu(MeCN)₄]BF₄ (3 mol%) catalysis. The latter showed high reactivity,

Table 1. Copper(I)-catalysed additions of ZnEt₂ to allylic halides **3–4**^a

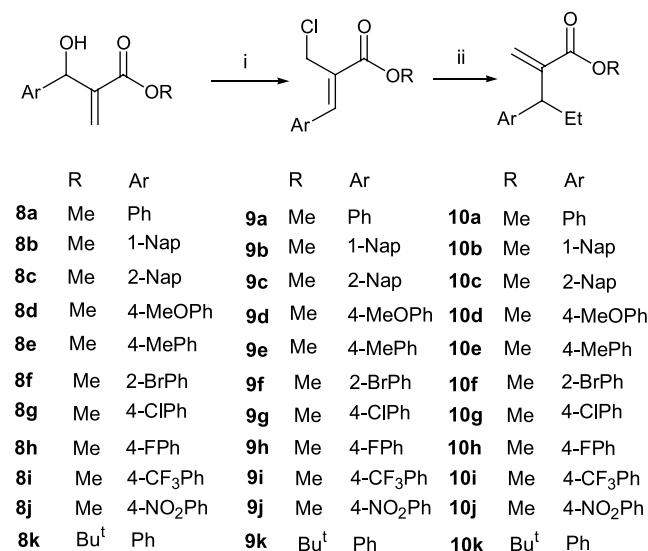
Halide	X	Ar	Yield 5 /%
3a	Br	Ph	100 (80)
3b	Br	1-Nap	84
3c	Br	4-MePh	86
3d	Br	4-ClPh	100 (63)
3e	Br	4-NO ₂ Ph	100 (63)
3f	Br	3-NO ₂ Ph	100 (80)
3g	Br	2-NO ₂ Ph	100 (50)
3h	Br	4-CF ₃ Ph	100 (95)
4a	Cl	Ph	100 (88)
4b	Cl	1-Nap	100 (95)
4c	Cl	4-MePh	100 (93)
4d	Cl	4-ClPh	100 (97)
4e	Cl	4-NO ₂ Ph	100 (97)
4f	Cl	3-NO ₂ Ph	100 (95)
4g	Cl	2-NO ₂ Ph	100 (98)
4h	Cl	4-CF ₃ Ph	100 (98)
6a	C(=O)H	Ph	0
7a	OMs	Ph	96

^a Addition of ZnEt₂ (1.5 equiv) to **3** or **4** (0.5–1.0 M in THF) at –20 °C in the presence of [Cu(MeCN)₄]BF₄ (3 mol%), followed by 40 min at –20 °C [for **3a–h**, **6a**, **7a**] or, cat. CuCN (20 mol%) and NBu₄Br (10 mol%) [for **4a–h**] (45 min, –20 °C).

^b Yields based on NMR or GC versus internal standards. Isolated yields in parentheses.

full conversion was attained even at -90°C within 5 h. These reactions are summarised in Table 1.

We had selected the products **5** and **10** as we felt the significant steric difference between the ethyl and phenyl groups would favour rapid separation of their enantiomers by automated chiral GC. In the end we could only identify a suitable GC column for separation of $(\pm)$ -**5a**. Separation of $(\pm)$ -**5b–c** and $(\pm)$ -**5e** was achieved on a Daicel OD HPLC column, but the enantiomers of compounds **5d** and **5f–h** could not be resolved by chiral HPLC. Details of the assay conditions are given in the experimental section (Table 4). Unfortunately, even in the favourable cases these HPLC assays are complicated in cases where less than a quantitative conversion to **5** is attained. The starting halides, **3** and **4** commonly co-elute with one of the enantiomers of **5** in the HPLC invalidating the assay. This problem can, however, be conveniently overcome by treatment of the crude reaction products with DABCO and subsequent washing with dilute acid leading to removal of **3**, **4** and excess DABCO as water soluble ammonium species. Other esters were investigated to extend the scope of the reaction and provide alternative systems for simpler enantiomeric assay. The methyl and *tert*-butyl esters **9** were prepared from **8** via use of oxalylchloride in the presence of a catalytic amount of DMF (Scheme 3). These reactions were entirely regiospecific under the conditions employed. The mildness of this approach was vital to the preparation of the *tert*-butyl ester **9i**. Attempts to form **9i** through the use of $\text{HCl}/\text{H}_2\text{SO}_4$ mixtures on **8i** led only to the derived free acid via *tert*-Bu cleavage. Use of the alternative catalyst system [CuCN (20 mol%) and TBAB (10 mol%)] allowed all of the β,β -disubstituted α -methylene propionates **10** to be isolated in good yield (Table 2). The enantiomers of the following methyl and *tert*-butyl esters could be separated by chiral GC: $(\pm)$ -**10a**, **10i** and **10k**. Compounds $(\pm)$ -**10b**, **10d–f**, **10h** and **10j** could be separated by chiral HPLC.



Scheme 3. Reagents and conditions: (i) $(\text{COCl})_2$ cat. DMF; (ii) ZnEt_2 , THF, -20°C , cat. CuCN (20 mol%) and NBu_4Br (10 mol%).

Table 2. Copper(I)-catalysed additions of ZnEt_2 to allylic chlorides **9**^a

Chloride	R	Ar	Yield 10 /%
9a	Me	Ph	95
9b	Me	1-Nap	98
9c	Me	2-Nap	94
9d	Me	4-MeOPh	91
9e	Me	4-MePh	97
9f	Me	2-BrPh	90
9g	Me	4-ClPh	95
9h	Me	4-FPh	94
9i	Me	4-CF ₃ Ph	85
9j	Me	4-NO ₂ Ph	97
9k	Bu ^t	Ph	83

^a Addition of ZnEt_2 (1.5 equiv) to **9** (0.5–1.0 M in THF) at -20°C in the presence of cat. CuCN (20 mol%) and NBu_4Br (10 mol%) followed by 40 min. At -20°C .

^b Isolated yields.

2.2. Scope of the organometallic reagent

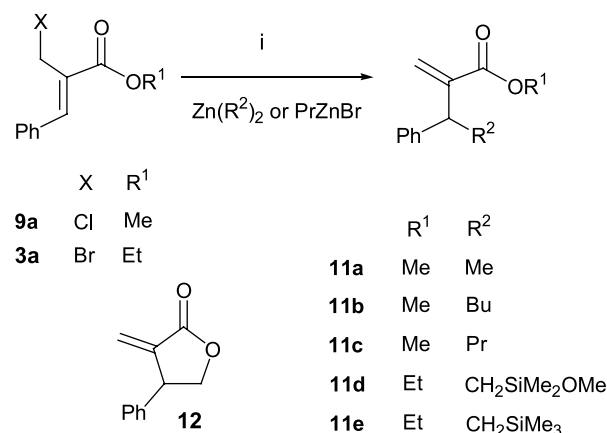
To increase the scope of the reaction beyond ZnEt_2 other organometallic reagents were investigated (Scheme 3). Both ZnMe_2 and ZnBu_2 participated in the catalytic reaction leading to good yields of isolated acrylates **11a–b** (Table 3). As the presumed by-products from reaction of ZnR_2 with **3**, **4** or **9** are RZnX ($\text{X} = \text{Br}, \text{Cl}$) we were encouraged to try the reaction with a representative organozinc halide. Rewardingly *n*-PrZnBr participated in the reaction almost as well as the diorganozincs. Attempts to extend this to use of Grignard reagents were unsuccessful though—complex

Table 3. Copper(I)-catalysed additions of organozincs to allylic halides **3** and **9**^a

Product	R ¹	R ²	Yield 11 /%
11a	Me	Me	85
11b	Me	Bu	98
11c	Me	Pr	70
11d	Et	$\text{CH}_2\text{SiMe}_2\text{OMe}$	93
11e	Et	CH_2SiMe_3	25

^a Addition of $\text{Zn}(\text{R}^2)_2$ (1.5 equiv) (0.5–1.0 M in THF) at -20°C in the presence of cat. CuCN (20 mol%) and NBu_4Br (10 mol%), followed by 40 min at -20°C for **11a–c**. Reaction -20 to -5°C , $\text{Zn}(\text{R}^2)_2$ (1.5 equiv), THF; $[\text{CuMeCN}]_4\text{BF}_4$ 3 mol% for **11d–e**.

^b Isolated yields.



Scheme 4. Reagents and conditions: (i) -20 to -5°C , THF; $[\text{CuMeCN}]_4\text{BF}_4$ 3 mol% or, cat. CuCN (20 mol%) and NBu_4Br (10 mol%) (40 min -20°C).

mixtures resulted presumably through attack at the carbonyl function of **11**. Some additional screenings were carried out using in situ prepared zinc reagents. Solutions of $\text{Zn}(\text{CH}_2\text{SiMe}_2\text{OMe})_2$, prepared by transmetallation of $\text{LiCH}_2\text{SiMe}_2\text{OMe}^{10}$ with ZnCl_2 . As expected this material demonstrated lower reactivity than the other organozinc species used but could be added to more reactive **3a** and expected product **11d** was isolated in acceptable yield (Scheme 4) provided an acidic workup is excluded. If this precaution is not undertaken **11d** dimerises via Si–O–Si bridges. The equivalent CH_2TMS derivative **11e** could also be formed, albeit in lower yield, through use of $\text{Zn}(\text{CH}_2\text{TMS})_2$. For **11d–e** all attempts to oxidatively cleave the Si–C bonds and fashion the known molluscicide **12** have been unsuccessful to date, either no reaction or complete oxidative degradation are observed.

3. Conclusions

We have identified a succinct route to β,β -disubstituted α -methylenepropionates that proceeds from simple allylic halide precursors attainable in multi-gram scales without the need for chromatography. The reactions of these with either diorganozincs or organozinc halides are chemo- and regiospecific and afford high yields of the desired products. The crude propionate products are of sufficient quality for immediate further use. While our studies have concentrated on allylic halides derived from aryl aldehydes this has been for convenience rather than due to a limitation in the synthetic method. It is worth noting that compound **5a** is a known intermediate in the synthesis of a potent ACE inhibitor (see experimental section). Synthesis of all of the aryl racemates **1** required for our further asymmetric catalytic studies has been attained and conditions developed for enantiomer assay on most of these species.

4. Experimental

4.1. General

All addition reactions (both stoichiometric and catalytic) involving air-sensitive reagents were performed in flame-dried Schlenk tubes under argon. All other preparative reactions were performed in a fumehood with appropriate precautions. Proton NMR spectra were recorded on a Bruker AM400 or AV400 (400 MHz) instruments; ^{13}C NMR were recorded on a Jeol EX270 or Bruker AV400 machines [all coupling constants (J) are given in Hz]. Mass spectra were recorded using electron ionisation (EI) and fast atom bombardment (FAB) or chemical ionisation (CH_4 , CI) techniques using Micromass 70E (EI) and Micromass Autospec (FAB) machines. Infrared spectra were recorded on a Perkin Elmer 1600 Series FTIR machine.

Various synthetic routes to the Baylis–Hillman adducts **2** and **8** are available.¹¹ All of these are well detailed in the literature except **2b** and **2f**. Some of the allylic bromides **3**¹² and chlorides (**4** and **9**)¹³ and the precursors to **7** are also known. All other compounds were commercial products or employed procedures described by us previously.¹⁴

4.2. Representative preparation of the Baylis–Hillman alcohols **2** and **8**

Neat aldehyde (20.0 mmol) was added to a stirred solution of ethyl acrylate (3.00 g, 30.0 mmol) and DABCO (2.23 g, 20.0 mmol) and left to stir overnight at room temperature. Reaction was monitored by TLC analysis (1:1 light petroleum/diethyl ether) until complete (1–30 days; electron-rich aromatic substrates react the slowest and were often further promoted by addition of lanthanum triflate, 10 mol%). The reaction mixture was extracted with diethyl ether (3×20 ml) and washed successively with HCl (2 M, 10 ml), NaOH (2 M, 10 ml) and brine (20 ml). The organic layer was separated and dried (MgSO_4) and the solvent removed to yield the crude oil. Purification by column chromatography (2:1 dichloromethane/light petroleum) gave the alcohols as oils.

4.2.1. Ethyl 2-(1-hydroxy-1-(1-naphthyl)-methyl)-propenoate 2b. Yield 75%; δ_{H} (400 MHz, CDCl_3) 1.17 (3H, t, $J=7.1$ Hz, CH_2Me), 3.05 (1H, d, $J=5.0$ Hz, CH), 4.12–4.17 (2H, m, CH_2Me), 5.48 (1H, s, $=\text{CH}_{2\alpha}$), 6.26 (1H, s, $=\text{CH}_{2\beta}$), 6.29 (1H, d, $J=5.0$ Hz, OH), 7.37–7.42 (3H, m, Ar), 7.56 (1H, d, $J=7.1$ Hz, Ar), 7.73 (1H, d, $J=8.2$ Hz, Ar), 7.77–7.80 (1H, m, Ar), 7.91–7.95 (1H, m, Ar); δ_{C} (100.6 MHz, CDCl_3) 13.9, 60.4, 69.0, 123.7, 124.4, 125.7, 125.9, 126.1, 126.6, 128.4, 128.6, 130.8, 133.7, 136.6, 142.2, 166.9; ν_{max} (CHCl_3 solution)/ cm^{-1} 3426br, 3048m, 2935s, 2903s, 1714s, 1629m, 1598m, 1510s, 1444m, 1148m, 1080m, 863m; m/z (EI) 256 (M^+ , 98%), 210 (63), 183 (56), 165 (84), 129 (100). [Found (HRMS, EI): M^+ 256.1101. $\text{C}_{16}\text{H}_{16}\text{O}_3$ requires M 256.1099].

4.2.2. Ethyl 2-(1-hydroxy-1-(3-nitrophenyl)-methyl)-propenoate 2f. Yield 92%; δ_{H} (400 MHz, CDCl_3) 1.28 (3H, t, $J=7.1$ Hz, CH_2Me), 3.35 (1H, br, OH), 4.22 (2H, q, $J=7.1$ Hz, CH_2Me), 5.64 (1H, s, br, unresolved long range coupling to $=\text{CH}_2$, ArCH), 5.90 (1H, s, $=\text{CH}_2$), 6.43 (1H, s, $=\text{CH}_2$), 7.54 (1H, t, $J=8.0$ Hz, Ar5-H), 7.78 (1H, m, Ar6-H), 8.18 (1H, m, Ar4-H), 8.28 (1H, apparent t, $J \sim 0.8$ Hz, Ar2-H); δ_{C} (67.8 MHz, CDCl_3) 14.0, 61.3, 72.6, 121.5, 122.7, 127.0, 129.3, 132.6, 141.2, 143.7, 148.2, 165.9; ν_{max} (CHCl_3 solution)/ cm^{-1} 3600, 2983m, 2939m, 1715s, 1629m, 1464m, 1372m, 1352s, 1330m, 1147m, 1095m, 1047m, 967m; m/z (EI) 251 (M^+ , 3%), 105 (22), 233 (47), 188 (56), 234 (100). [Found (HRMS, EI): M^+ 251.0803. $\text{C}_{12}\text{H}_{13}\text{NO}_5$ requires M 251.0794].

4.3. General preparation of the allylic halides **3–4**

Either concentrated hydrobromic acid (4.2 ml of a 48% w/w solution, 25.0 mmol; for **3**) [or concentrated HCl (2.5 ml of 37% w/w solution, 25.0 mmol; for **4**)] followed by concentrated sulfuric acid (3.8 ml of a 98% w/w solution, 38 mmol; for **3**) [or (2.3 ml of a 98% w/w solution, 23 mmol; for **4**)] were added dropwise to stirred Baylis–Hillman alcohol (25.0 mmol) using an ice bath for cooling. The mixture was stirred overnight at room temperature and extracted with diethyl ether (2×20 ml). The organic layers were combined and washed successively with saturated sodium hydrogen carbonate solution (10 ml), distilled water (10 ml) and brine (10 ml), dried (Na_2SO_4) and the solvent

removed to give a yellow oils which were crystallised from pentane or Kugelrohr distilled.

4.3.1. Ethyl (Z)-2-bromomethyl-3-(1-naphthyl)-propenoate 3b. Yield 62%; low melting point solid; δ_{H} (400 MHz, CDCl_3) 1.45 (3H, t, $J=7.2$ Hz, CH_2Me), 4.34 (2H, s, CH_2Br), 4.43 (2H, q, $J=7.2$ Hz, CH_2Me), 7.54–7.61 (3H, m, Ar), 7.79–7.94 (4H, m, Ar), 8.34 (1H, s, $=\text{CH}$); δ_{C} (67.8 MHz, CDCl_3) 14.4, 27.0, 61.6, 124.2, 125.4, 126.2, 126.5, 126.8, 128.7, 129.8, 131.2, 131.6, 133.5, 141.0, 165.9; ν_{max} (CHCl_3 solution)/ cm^{-1} 2978m, 1715s, 1628m, 1366m, 1277s, 1256s, 1151m, 1089m, 1021m, 805s; m/z (FAB) 318 (M^+ , ^{79}Br , 10%), 240 (22), 239 (100). [Found (HRMS, FAB): M^+ , 318.0265. $\text{C}_{16}\text{H}_{15}\text{BrO}_2$ requires M 318.0255].

4.3.2. Ethyl (Z)-2-chloromethyl-3-(1-naphthyl)-propenoate 4b. Yield 74%; low melting solid; δ_{H} (400 MHz, CDCl_3) 1.44 (3H, t, $J=7.1$ Hz, CH_2Me), 4.42 (4H, s coincident with q, $J=7.1$ Hz, CH_2Cl and CH_2Me), 7.54–7.58 (3H, m, Ar), 7.73 (1H, d, $J=7.1$ Hz, Ar), 7.89–7.93 (3H, m, Ar), 8.43 (1H, s, $=\text{CH}$); δ_{C} (100.6 MHz, CDCl_3) 14.8, 39.9, 61.9, 124.7, 125.4, 126.9, 127.1, 127.3, 129.0, 130.4, 131.8, 131.9, 133.9, 138.6, 143.9, 166.2; ν_{max} (CHCl_3 solution)/ cm^{-1} 2980m, 1715s, 1629m, 1369m, 1341m, 1255s, 1185s, 1098m, 805m, 778s; m/z (FAB) 274 (M^+ , ^{35}Cl , 68%), 240 (19), 239 (68), 209, 195 (40), 167 (48), 166 (40), 165 (100). [Found (HRMS, FAB): M^+ , 274.0765. $\text{C}_{16}\text{H}_{15}\text{ClO}_2$ requires M 274.0761].

4.3.3. Ethyl (Z)-2-chloromethyl-3-(3-nitrophenyl)-propenoate 4f. Yield 47%; mp 55 °C (pentane) (Found: C, 53.5; H, 4.4; N, 5.3%. Calcd for $\text{C}_{12}\text{H}_{12}\text{NO}_4\text{Cl}$: C, 53.4; H, 4.5; N, 5.2%); δ_{H} (400 MHz, CDCl_3) 1.41 (3, t, $J=7.1$ Hz, CH_2Me), 4.38 (2H, q, $J=7.1$ Hz, CH_2Me), 4.43 (2H, s, CH_2Cl), 7.65 (1H, apparent t, $J=8.4$ Hz, Ar), 7.86 (1H, s, $=\text{CH}$), 7.89 (1H, ddd, $J=8.4, 1.3, 1.0$ Hz, Ar), 8.21 (1H, dd, $J=8.4, 1.3, 1.0$ Hz, Ar), 8.40 (1H, dd, $J=1.3, 1.0$ Hz, Ar); δ_{C} (67.8 MHz, CDCl_3) 14.1, 38.1, 61.7, 124.0, 124.2, 130.0, 131.1, 134.9, 135.7, 140.2, 148.4, 165.4; ν_{max} (CHCl_3 solution)/ cm^{-1} 2898m, 1716s, 1638m, 1359s, 1300s, 1093s, 1021m, 897w, 875m; m/z (EI) 269 (M^+ , ^{35}Cl , 38%), 234 (36), 224 (39), 161 (22), 149 (20), 116 (22), 115 (100). [Found (HRMS, EI): M^+ , 269.0443. $\text{C}_{12}\text{H}_{12}\text{ClNO}_4$ requires M 269.0455].

4.3.4. Ethyl (Z)-2-chloromethyl-3-(4-trifluoromethyl-phenyl)-propenoate 4h. Yield 86%; low melting solid. (Found: C, 53.1; H, 4.0%. Calcd for $\text{C}_{13}\text{H}_{12}\text{ClF}_3\text{O}_2$: C, 53.35; H, 4.1%); δ_{H} (400 MHz, CDCl_3) 1.40 (3H, t, $J=7.1$ Hz, CH_2Me), 4.36 (2H, q, $J=7.1$ Hz, CH_2Me), 4.42 (2H, s, CH_2Cl), 7.66 (2H, d, $J=8.3$ Hz, Ar), 7.73 (2H, d, $J=8.3$ Hz, Ar), 7.87 (1H, s, $=\text{CH}$); δ_{C} (100.6 MHz, CDCl_3) 14.2, 38.5, 61.5, 123.9 (q, $^1J_{\text{CF}}=272$ Hz), 125.7 (q, $^3J_{\text{CF}}=4$ Hz), 129.7, 131.0, 131.2 (q, $^2J_{\text{CF}}=33$ Hz), 137.8, 141.6, 165.8; ν_{max} (CHCl_3 solution)/ cm^{-1} 2943w, 1713s, 1637m, 1373m, 1325s, 1301m, 1284m, 1168m, 1132s, 1115m, 1095m, 1068s, 1018m; m/z (EI) 292 (M^+ , 79%), 247 (54), 218 (19), 183 (100). [Found (HRMS, EI): M^+ 292.0477. $\text{C}_{13}\text{H}_{12}\text{O}_2\text{F}_3\text{Cl}$ requires M 292.0478].

4.4. Representative addition of organometallics to the allylic halides 3–4 and 9

A solution of ZnEt_2 (1.5 ml of 1.0 M hexane solution, 1.5 mmol) was added to a cold (-20°C) solution of **3** (1.00 mmol) and $[\text{Cu}(\text{MeCN})_4]\text{BF}_4$ (9.4 mg, 0.03 mmol) in THF (1.0 ml). The reaction was stirred (40 min) at -20°C and the reaction quenched with 2 M HCl_{aq} . The reaction mixture was partitioned between dichloromethane and water, the organic layer separated, dried (MgSO_4), and the solvent removed to give essentially pure **5**. Similar reactions were carried out using **4** and **9** (1.00 mmol) but with a catalyst system comprised of CuCN (17.9 mg, 0.2 mmol) and TBAB (32 mg, 0.1 mmol) in THF (1.0 ml). The reaction was stirred for 25 min at -20°C and the reaction was quenched with 2 M HCl . Similar procedures were used for the additions of other ZnR_2 and $n\text{-PrZnBr}$ reagents.

Alternative workup to facilitate removal of unreacted starting halides 3–4. The crude oil (140 mg) was dissolved in diethyl ether (10 ml) and DABCO (80 mg, 0.75 mmol) was added. The solution was stirred overnight at room temperature and then transferred to a separating funnel where it was washed with HCl (1 M, 5 ml) and brine (10 ml), dried (Na_2SO_4) and the solvent evaporated to give a colourless oil. If analysis by ^1H NMR spectroscopy showed that there was still a residual amount of halide (especially for **4**) the oil was re-dissolved in diethyl ether (5 ml), DABCO (60 mg, 0.54 mmol) was added and the solution was stirred overnight at room temperature again. After washing and drying as before the oil was found to be solely $\text{S}_{\text{N}}2'$ product **5**.

4.4.1. Ethyl 2-methylene-3-phenyl-pentanoate 5a. Yield 88%; δ_{H} (400 MHz, CDCl_3) 0.86 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.20 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.77 (1H, ddq, $J=13.4, 8.9, 7.3$ Hz, $\text{CHCH}_2\alpha\text{Me}$), 1.89 (1H, ddq, $J=13.4, 6.3, 7.3$ Hz, $\text{CHCH}_2\beta\text{Me}$), 3.73 (1H, dd, $J=8.9, 6.3$ Hz, CH), 4.09 (2H, m, OCH_2Me), 5.63 (2H, s, $=\text{CH}_2\alpha$), 6.29 (1H, s, $=\text{CH}_2\beta$), 7.16–7.29 (5H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.4, 14.0, 27.4, 48.0, 60.6, 123.4, 126.2, 128.1, 128.2, 142.8, 144.1, 167.1; ν_{max} (thin film)/ cm^{-1} 2964m, 1718s, 1627m, 1452m, 1368m, 1252s, 1149s; m/z (EI) 218 (M^+ , 100%). [Found (HRMS, EI): M^+ 218.1302. $\text{C}_{14}\text{H}_{18}\text{O}_2$ requires M 218.1307]. This compound has been prepared once by an alternative route.¹⁵

4.4.2. Ethyl 2-methylene-3-(1-naphthyl)-pentanoate 5b. Yield 95%; δ_{H} (400 MHz, CDCl_3) 0.96 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.14 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.92–2.06 (2H, m, CHCH_2Me), 4.09–4.17 (2H, m, OCH_2Me), 4.71 (1H, dd, $J=7.4, 7.2$ Hz, CH), 5.56 (1H, s, $=\text{CH}_2\alpha$), 6.33 (1H, s, $=\text{CH}_2\beta$), 7.38 (1H, dd, $J=7.2, 1.1$ Hz, Ar), 7.42–7.55 (3H, m, Ar), 7.73 (1H, d, $J=8.1$ Hz, Ar), 7.85 (1H, dd, $J=8.0, 1.4$ Hz, Ar), 8.19 (1H, d, $J=8.5$ Hz, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.8, 14.2, 28.2, 42.2, 48.3, 60.9, 123.9, 124.2, 124.5, 125.4, 125.5, 126.0, 128.9, 132.4, 134.2, 139.3, 144.6, 167.6; ν_{max} (thin film)/ cm^{-1} 2960m, 2927m, 1713s, 1251m, 1145m, 797m; m/z (FAB) 268 (M^+ , 37%), 223 (69), 207 (51), 195 (37), 147 (100). [Found (HRMS, FAB): M^+ , 268.1456. $\text{C}_{18}\text{H}_{20}\text{O}_2$ requires M 268.1463].

4.4.3. Ethyl 2-methylene-3-(4-tolyl)-pentanoate 5c. Yield 93%; δ_{H} (400 MHz, CDCl_3) 0.76 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.11 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.64 (1H, ddq, $J=13.4, 8.6, 7.3$ Hz, $\text{CHCH}_{2\alpha}\text{Me}$), 1.78 (1H, ddq, $J=13.4, 6.5, 7.3$ Hz, $\text{CHCH}_{2\beta}\text{Me}$), 2.20 (3H, s, ArMe), 3.61 (1H, dd, $J=8.6, 6.5$ Hz, CH), 4.00 (2H, m, OCH_2Me), 5.51 (1H, s, $=\text{CH}_{2\alpha}$), 6.17 (1H, s, $=\text{CH}_{2\beta}$), 6.97–7.01 (4H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.3, 14.0, 20.9, 27.4, 47.5, 60.5, 123.1, 127.9, 128.8, 129.1, 139.7, 144.3, 167.3; ν_{max} (thin film)/ cm^{-1} 2955m, 1717s, 1628m, 1512m, 1252m, 1149m, 814m; m/z (CI) 233 ($[\text{M}+\text{H}]^+$, 100%). [Found (HRMS, CI): $[\text{M}+\text{H}]^+$, 233.1542. $\text{C}_{15}\text{H}_{21}\text{O}_2$ requires $[\text{M}+\text{H}]$ 233.1542].

4.4.4. Ethyl 3-(4-chlorophenyl)-2-methylene-pentanoate 5d. Yield 97%; δ_{H} (400 MHz, CDCl_3) 0.84 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.19 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.71 (1H, ddq, $J=13.5, 8.8, 7.3$ Hz, $\text{CHCH}_{2\alpha}\text{Me}$), 1.87 (1H, ddq, $J=13.5, 6.3, 7.3$ Hz, $\text{CHCH}_{2\beta}\text{Me}$), 3.89 (1H, dd, $J=8.8, 6.3$ Hz, CH), 4.11 (2H, m, OCH_2Me), 5.63 (1H, s, $=\text{CH}_{2\alpha}$), 6.30 (1H, s, $=\text{CH}_{2\beta}$), 7.13 (2H, d, $J=8.4$ Hz, Ar), 7.23 (2H, d, $J=8.4$ Hz, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.1, 13.9, 27.2, 47.3, 60.5, 123.5, 128.2, 129.4, 131.8, 141.3, 143.6, 166.6; ν_{max} (thin film)/ cm^{-1} 2965m, 1717s, 1627m, 1491m, 1252s, 1151s, 1092s, 1015m, 820m; m/z (FAB) 252 (M^+ , ^{35}Cl , 12%), 219 (22), 207 (30), 176 (22), 155 (50), 154 (100). [Found (HRMS, FAB): M^+ , 252.0912. $\text{C}_{14}\text{H}_{17}\text{ClO}_2$ requires M 252.0917].

4.4.5. Ethyl 2-methylene-3-(4-nitrophenyl)-pentanoate 5e. Yield 97%; δ_{H} (400 MHz, CDCl_3) 0.88 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.21 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.79 (1H, ddq, $J=13.4, 8.8, 7.3$ Hz, $\text{CHCH}_{2\alpha}\text{Me}$), 1.78 (1H, ddq, $J=13.4, 6.3, 7.3$ Hz, $\text{CHCH}_{2\beta}\text{Me}$), 3.84 (1H, dd, $J=8.8, 6.3$ Hz, CH), 4.11 (2H, m, OCH_2Me), 5.76 (1H, s, $=\text{CH}_{2\alpha}$), 6.41 (1H, d, $=\text{CH}_{2\beta}$), 7.39 (2H, d, $J=8.7$ Hz, Ar), 8.14 (2H, d, $J=8.7$ Hz, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.0, 13.8, 26.9, 47.8, 60.6, 123.3, 124.5, 128.8, 142.4, 146.3, 150.7, 166.2; ν_{max} (thin film)/ cm^{-1} 2967m, 1714s ($\text{C}=\text{O}$), 1520s, 1347m, 1254m, 1152m, 851m; m/z (FAB) 264 ($[\text{M}+\text{H}]^+$, 13%), 221 (14), 207 (16), 147 (38), 77 (13), 73 (100). [Found (HRMS, FAB): $[\text{M}+\text{H}]^+$, 264.1244. $\text{C}_{14}\text{H}_{18}\text{NO}_4$ requires $[\text{M}+\text{H}]$ 264.1236].

4.4.6. Ethyl 2-methylene-3-(3-nitrophenyl)-pentanoate 5f. Yield 95%; δ_{H} (400 MHz, CDCl_3) 0.88 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.22 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.82 (1H, ddq, $J=13.8, 8.8, 7.3$ Hz, $\text{CHCH}_{2\alpha}\text{Me}$), 1.94 (1H, ddq, $J=13.7, 6.6, 7.3$ Hz, $\text{CHCH}_{2\beta}\text{Me}$), 3.86 (1H, dd, $J=7.3, 8.8$ Hz, CH), 4.07–4.16 (2H, m, OCH_2Me), 5.76 (1H, s, $=\text{CH}_{2\alpha}$), 6.38 (1H, s, $=\text{CH}_{2\beta}$), 7.45 (1H, ddd, $J=7.5, 8.1, 1.5$ Hz, Ar), 7.57 (1H, dt, $J=7.5, 1.0$ Hz, Ar), 8.06–8.07 (1H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.5, 14.3, 27.5, 48.1, 61.2, 121.9, 123.1, 125.0, 129.4, 134.9, 143.0, 145.6, 148.6, 166.7; ν_{max} (CHCl_3 solution)/ cm^{-1} 2995m, 1710s, 1629m, 1462m, 1353s, 1097m, 1025m, 954m, 900s, 648m; m/z (EI) 263 (M^+ , 32%), 246 (60), 218 (64), 172 (83), 143 (63), 115 (100). [Found (HRMS, EI): M^+ 263.1158. $\text{C}_{14}\text{H}_{17}\text{O}_4\text{N}$ requires M 263.1157].

4.4.7. Ethyl 2-methylene-3-(2-nitrophenyl)-pentanoate 5g. Yield 98%; δ_{H} (400 MHz, CDCl_3) 0.90 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.06 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.83

(1H, ddq, $J=14.7, 7.3, 6.4$ Hz, $\text{CHCH}_{2\alpha}\text{Me}$), 1.89 (1H, ddq, $J=14.5, 7.3, 6.6$ Hz, $\text{CHCH}_{2\beta}\text{Me}$), 3.96–4.02 (2H, m, OCH_2Me), 4.32 (1H, t, $J=7.3$ Hz, CH), 5.66 (1H, s, $=\text{CH}_{2\alpha}$), 6.34 (1H, s, $=\text{CH}_{2\beta}$), 7.22–7.27 (2H, m, Ar), 7.43 (1H, ddd, $J=8.7, 7.6, 1$ Hz, Ar), 7.72 (1H, dd, $J=8.1$ Hz, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.2, 14.1, 27.4, 41.6, 61.1, 124.4, 124.9, 127.2, 129.0, 132.5, 137.8, 142.5, 150.6, 166.5; ν_{max} (thin film)/ cm^{-1} 2935m, 1714s, 1630m, 1578m, 1527s, 1463m, 1356s; m/z (EI) 263 (M^+ , 1%), 255 (36), 218 (100). [Found (HRMS, EI): M^+ 263.1168. $\text{C}_{14}\text{H}_{17}\text{O}_4\text{N}$ requires M 263.1157].

4.4.8. Ethyl 2-methylene-3-(4-trifluoromethylphenyl)-pentanoate 5h. Yield 98%; δ_{H} (400 MHz, CDCl_3) 0.86 (3H, t, $J=7.3$ Hz, CHCH_2Me), 1.21 (3H, t, $J=7.1$ Hz, OCH_2Me), 1.76 (1H, ddq, $J=14.3, 8.9, 7.3$ Hz, $\text{CHCH}_{2\alpha}\text{Me}$), 1.91 (1H, ddq, $J=14.3, 6.5, 7.3$ Hz, $\text{CHCH}_{2\beta}\text{Me}$), 3.80 (1H, dd, $J=8.9, 6.6$ Hz, CH_2), 4.02–4.19 (2H, m, OCH_2Me), 5.70 (1H, s, $=\text{CH}_{2\alpha}$), 6.37 (1H, s, $=\text{CH}_{2\beta}$), 7.33 (2H, d, $J=8.2$ Hz, Ar), 7.54 (2H, d, $J=8.2$ Hz, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.4, 14.3, 27.4, 48.1, 60.9, 124.3, 124.5 (q, $^1J_{\text{CF}}=272$ Hz), 125.4 (q, $^3J_{\text{CF}}=3$ Hz), 128.6, 128.8 (q, $^2J_{\text{CF}}=32$ Hz), 143.5, 147.3, 166.9; ν_{max} (CHCl_3 solution)/ cm^{-1} 2933m, 1711s, 1618m, 1325s, 1125s, 1068s; m/z (EI) 286 (M^+ , 100%). [Found (HRMS, EI): M^+ 286.1175. $\text{C}_{15}\text{H}_{17}\text{O}_2\text{F}_3$ requires M 286.1181].

4.5. Preparation of ethyl (Z)-2-methanesulfonylmethyl-3-phenyl-propenoate 7

The formate **6** was prepared by a literature method and hydrolysed to the derived allylic alcohol.^{11f} Both compounds showed identical spectroscopic and physical properties to those published. Mesyl chloride (0.4 ml, 5.1 mmol) was added to a stirred solution of allylic alcohol (0.7 g, 3.4 mmol) in dichloromethane (10 ml) at 0 °C, followed by dropwise addition of NEt_3 (0.96 ml, 6.9 mmol). The reaction mixture was stirred at 0 °C for 30 min. The solution was diluted with diethyl ether (10 ml), poured onto ice-water and transferred to a separating funnel. The organic layer was washed with chilled (0 °C) HCl (2 M, 5 ml) and chilled (0 °C) brine (10 ml). The aqueous layer was re-extracted with ether (10 ml) and washed as before. The combined extracts were dried (Na_2SO_4) and the solvent evaporated to give the crude oil which crystallised overnight at –20 °C to give **7** (0.81 g, 84%) as a yellow solid. No further purification was attempted due to the reactive nature of **7a**, however, the spectroscopic spectrum showed the product to be essentially pure.

Mp dec. slowly at ambient temperature ($t_{1/2} \sim 10$ h); δ_{H} (400 MHz, CDCl_3) 1.38 (3H, t, $J=7.1$ Hz, OCH_2Me), 3.12 (3H, s, $-\text{SO}_2\text{Me}$), 4.33 (2H, q, $J=7.1$ Hz, OCH_2Me), 5.07 (2H, s, CH_2OMs), 7.43–7.48 (5H, m, Ar), 8.08 (1H, s, $=\text{CH}$); δ_{C} (67.8 MHz, CDCl_3) 14.3, 37.4, 61.5, 65.1, 124.4, 128.9 (2C), 129.5 (2C), 130.2, 133.5, 147.6, 166.3; ν_{max} (CHCl_3 solution)/ cm^{-1} 3027m, 2980m, 2934m, 1709s, 1631s, 1448m, 1358s, 1347s, 1323m, 1241s, 1175s; m/z (EI) 284 (M^+ , 9%), 189 (13), 188 (13), 159 (64), 115 (100). [Found (HRMS, EI): M^+ 284.0715. $\text{C}_{13}\text{H}_{16}\text{O}_5\text{S}$ requires M 284.0718].

4.6. Representative preparation of allylic chlorides 9

Oxalyl chloride (0.5 ml, 5.7 mmol) was added dropwise over 2 min to a stirred solution of allylic alcohol (4.0 mmol) and *N,N*-dimethylformamide (0.1 ml; 33 mol%) in chloroform (10 ml) at room temperature under argon. After stirring for a 3 h the reaction was quenched by cautious addition of 2 M NaOH_(aq.) (10 ml). After dilution with EtOAc (50 ml) and H₂O (50 ml) the organic layer was separated, dried (MgSO₄) and concentrated. Flash chromatography (silica; EtOAc/Hexanes) gave the products as colourless oils and solids.

4.6.1. Methyl (Z)-2-chloromethyl-3-(1-naphthyl)-propenoate 9b. Yield 54%; mp 74–76 °C (Heptane). (Found: C, 69.07; H, 5.04%. Calcd for C₁₅H₁₃ClO₂: C, 69.10; H, 5.03%); δ_{H} (400 MHz, CDCl₃) 3.95 (3H, s, OMe), 4.41 (2H, s, CH₂Cl), 7.53–7.57 (3H, m, Ar), 7.72 (1H, dt, *J* = 7.1, 1.0 Hz, Ar), 7.88–7.92 (3H, m, Ar), 8.43 (1H, s, =CH); δ_{C} (100.6 MHz, CDCl₃) 39.6, 52.7, 124.4, 125.6, 126.6, 126.9, 127.0, 128.9, 130.1, 130.7, 131.5, 131.6, 133.6, 142.2, 166.6; ν_{max} (CHCl₃ solution)/cm⁻¹ 2954w, 1716s, 1634w, 1343m, 1289m, 1099m, 977w; *m/z* (EI) 260 (M⁺, ³⁵Cl, 65%), 201 (22), 193 (20), 165 (100). [Found (HRMS, EI) M⁺ 260.0613. C₁₅H₁₃O₂Cl requires M 260.0604].

4.6.2. Methyl (Z)-2-chloromethyl-3-(2-naphthyl)-propenoate 9c. Yield 67%; mp 76–78 °C (Heptane); δ_{H} (400 MHz, CDCl₃) 3.91 (3H, s, OMe), 4.56 (2H, s, CH₂Cl), 7.51–7.58 (2H, m, Ar), 7.62 (1H, dd, *J* = 8.6, 1.8 Hz, Ar), 7.85–7.92 (3H, m, Ar), 8.04 (1H, s, =CH), 8.08 (1H, d, *J* = 0.8 Hz, Ar); δ_{C} (100.6 MHz, CDCl₃) 39.5, 52.7, 126.6, 126.9, 127.6, 127.9, 128.6, 128.8, 128.9, 130.2, 131.8, 133.3, 133.7, 144.1, 167.0; ν_{max} (CHCl₃ solution)/cm⁻¹ 2953w, 1714s, 1624m, 1286m, 1092m, 861w; *m/z* (EI) 260 (M⁺, ³⁵Cl, 67%), 225 (26), 193 (11), 165 (100). [Found (HRMS, EI) M⁺ 260.0598. C₁₅H₁₃O₂Cl requires M 260.0604].

4.6.3. Methyl (Z)-2-chloromethyl-3-(4-tolyl)-propenoate 9e. Yield 67%. (Found: C, 64.16; H, 5.80%. Calcd for C₁₂H₁₃ClO₂: C, 64.15; H, 5.83%); δ_{H} (400 MHz, CDCl₃) 2.40 (3H, s, ArMe), 3.87 (3H, s, OMe), 4.50 (2H, s, CH₂Cl), 7.26 (2H, d, *J* = 8.1 Hz, Ar), 7.47 (2H, d, *J* = 8.1 Hz, Ar), 7.85 (1H, s, =CH); δ_{C} (67.8 MHz, CDCl₃) 21.6, 39.5, 52.6, 127.5, 129.8, 130.0, 131.5, 140.4, 144.2, 167.1; ν_{max} (thin film)/cm⁻¹ 2951m, 1714s, 1436s, 1091s, 928m; *m/z* (EI) 224 (M⁺, ³⁵Cl, 44%), 189 (66), 129 (100). [Found (HRMS, EI) M⁺ 224.0611. C₁₂H₁₃ClO₂ requires M 224.0604].

4.6.4. Methyl (Z)-2-chloromethyl-3-(2-bromophenyl)-propenoate 9f. Yield 82%; mp 73–74 °C. (Found: C, 45.50; H, 3.34%. Calcd for C₁₁H₁₀O₂BrCl: C, 45.63; H, 3.48%); δ_{H} (400 MHz, CDCl₃) 3.91 (3H, s, OMe), 4.34 (2H, s, CH₂Cl), 7.25–7.29 (1H, m, Ar), 7.40–7.44 (1H, m, Ar), 7.62–7.66 (2H, m, Ar), 7.90 (1H, s, =CH); δ_{C} (100.6 MHz, CDCl₃) 39.1, 52.8, 124.6, 127.8, 130.2, 130.3, 131.0, 133.2, 134.8, 142.7, 166.3; ν_{max} (CHCl₃ solution)/cm⁻¹ 2954w, 1716s, 1636m, 1364m, 1293m, 1094m; *m/z* (EI) 288 (M⁺, ³⁵Cl/⁷⁹Br, 2%), 257 (4), 209 (100). [Found (HRMS, EI) M⁺ 287.9567. C₁₁H₁₀O₂BrCl requires M 287.9553].

4.6.5. Methyl (Z)-2-chloromethyl-3-(4-chlorophenyl)-

propenoate 9f. Yield 83%; δ_{H} (400 MHz, CDCl₃) 3.88 (3H, s, OMe), 4.34 (2H, s, CH₂Cl), 7.42–7.45 (2H, m, Ar), 7.48–7.51 (2H, m, Ar), 7.82 (1H, s, =CH); δ_{C} (67.8 MHz, CDCl₃) 39.0, 52.8, 129.0, 129.4, 131.1, 132.7, 136.1, 142.6, 166.7; ν_{max} (thin film)/cm⁻¹ 2951m, 1716s, 1491s, 1436s, 1089s, 1014m, 928m; *m/z* (EI) 244 (M⁺, ³⁵Cl, 59%), 209 (80), 149 (100). [Found (HRMS, EI) M⁺ 244.0061. C₁₁H₁₁O₂Cl₂ requires M 244.0058].

4.6.6. Methyl (Z)-2-chloromethyl-3-(4-trifluoromethyl-phenyl)-propenoate 9i. Yield 69%; δ_{H} (400 MHz, CDCl₃) 3.90 (3H, s, OMe), 4.42 (2H, s, CH₂Cl), 7.65 (2H, d, *J* = 8.4 Hz, Ar), 7.72 (2H, d, *J* = 8.4 Hz, Ar), 7.88 (1H, s, =CH); δ_{C} (100.6 MHz, CDCl₃) 38.6, 52.8, 125.3 (q, ¹*J*_{CF} = 272.2 Hz), 126.0 (q, ³*J*_{CF} = 3.7 Hz), 129.8, 130.7, 131.5 (q, ²*J*_{CF} = 33.0 Hz), 137.8, 142.0, 166.4; ν_{max} (thin film)/cm⁻¹ 2955w, 1727s, 1635m, 1616m, 1437m, 1323s, 1281s, 1170s, 1127s, 1018m, 845m, 783m; *m/z* (EI) 278 (M⁺, ³⁵Cl, 58%), 259 (11), 218 (18), 183 (100). [Found (HRMS, EI) M⁺ 278.0329. C₁₂H₁₀O₂F₃Cl requires M 278.0321].

4.6.7. Methyl (Z)-2-chloromethyl-3-(4-nitrophenyl)-propenoate 9j. Yield 90%; mp 126–128 °C; δ_{H} (400 MHz, CDCl₃) 3.93 (3H, s, OMe), 4.42 (2H, s, CH₂Cl), 7.71–7.74 (2H, m, Ar), 7.90 (1H, s, =CH), 8.32–8.35 (2H, m, Ar); δ_{C} (100.6 MHz, CDCl₃) 38.3, 53.0, 124.3, 130.4, 131.8, 140.6, 140.9, 148.3, 166.1; ν_{max} (CHCl₃ solution)/cm⁻¹ 2954w, 1722s, 1598m, 1384s, 1092m, 981w; *m/z* (EI) 255 (M⁺, ³⁵Cl, 69%), 224 (34), 174 (78), 144 (27), 115 (100). [Found (HRMS, EI) M⁺ 255.0925. C₁₁H₁₀NO₄Cl requires M 255.0298].

4.6.8. tert-Butyl (Z)-2-chloromethyl-3-phenyl-propenoate 9k. Yield 80%. (Found: C, 66.84; H, 6.72%. Calcd for C₁₄H₁₇O₂Cl: C, 66.53; H, 6.78%); δ_{H} (400 MHz, CDCl₃) 1.57 (9H, s, *t*Bu), 4.43 (2H, s, CH₂Cl), 7.39–7.54 (5H, m, Ar), 7.78 (1H, s, =CH); δ_{C} (100.6 MHz, CDCl₃) 28.3, 39.5, 81.8, 128.7, 129.0 (2C), 129.6, 129.7 (2C), 134.7, 142.7, 165.5; ν_{max} (thin film)/cm⁻¹ 2934m, 1709s, 1632m, 1494m, 1451m, 1368s, 1155s, 1094m, 850m; *m/z* (EI) 252 (M⁺, ³⁵Cl, 11%), 196 (86), 179 (50), 115 (72), 57 (100). [Found (HRMS, EI) M⁺ 252.0930. C₁₄H₁₇O₂Cl requires M 252.0917].

4.7. General preparation of the β,β -disubstituted α -methylenepropionates 10 and 11a–c

These compounds were prepared in an analogous manner to compounds 5.

4.7.1. Methyl-2-methylene-3-phenyl-pentanoate 10a. Yield 95%; colourless oil. (Found: C, 76.18; H, 7.85%. Calcd for C₁₃H₁₆O₂: C, 76.44; H, 7.90%); δ_{H} (500 MHz, CDCl₃) 0.86 (3H, t, *J* = 7.3 Hz, CH₂Me), 1.72–1.93 (2H, m, CH₂Me), 3.67 (3H, s, OMe), 3.73 (1H, dd, *J* = 8.3, 7.0 Hz, CH), 5.65 (1H, s, =CH_{2 α}), 6.29 (1H, s, =CH_{2 β}), 7.17–7.29 (5H, m, Ar); δ_{C} (125 MHz, CDCl₃) 12.6, 27.7, 48.2, 52.0, 124.0, 126.5, 128.3 (2C), 128.4 (2C), 142.9, 144.0, 167.7; ν_{max} (thin film)/cm⁻¹ 2962m, 2875m, 1720s, 1623m, 1437m, 1255m, 1151m, 946m; *m/z* (EI) 204 (M⁺, 89%), 172 (53), 144 (48), 129 (31), 115 (100). [Found (HRMS, EI) M⁺ 204.1144. C₁₃H₁₆O₂ requires M 205.1150].

4.7.2. Methyl-2-methylene-3-(1-naphthyl)-pentanoate

10b. Yield 98%; δ_{H} (400 MHz, CDCl_3) 0.94 (3H, t, $J=7.3$ Hz, CH_2Me), 1.88–2.05 (2H, m, CH_2Me), 3.67 (3H, s, OMe), 4.69 (1H, br dd, $J=7.3$ Hz, unresolved, CH), 5.53 (1H, s, $=\text{CH}_{2\alpha}$), 6.30 (1H, s, $=\text{CH}_{2\beta}$), 7.35–7.53 (4H, m, Ar), 7.71 (1H, d, $J=8.1$ Hz, Ar), 7.83 (1H, dd, $J=8.0$, 1.5 Hz, Ar), 8.15 (1H, d, $J=8.4$ Hz, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.9, 28.2, 42.3, 52.1, 123.8, 124.2, 124.9, 125.4, 125.6, 126.1, 127.2, 128.9, 132.3, 134.2, 139.1, 144.2, 168.0; ν_{max} (thin film)/ cm^{-1} 2962m, 1721s, 1627m, 1427m, 1248s, 948m; m/z (EI) 254 (M^+ , 39%), 225 (44), 195 (39), 165 (100). [Found (HRMS, EI) M^+ 254.1306. $\text{C}_{17}\text{H}_{18}\text{O}_2$ requires M 254.1307].

4.7.3. Methyl-2-methylene-3-(2-naphthyl)-pentanoate

10c. Yield 94%; δ_{H} (400 MHz, CDCl_3) 0.89 (3H, t, $J=7.3$ Hz, CH_2Me), 1.82–2.02 (2H, m, CH_2Me), 3.65 (3H, s, OMe), 3.91 (1H, dd, $J=8.7$, 6.4 Hz, CH), 5.72 (1H, s, $=\text{CH}_{2\alpha}$), 6.34 (1H, s, $=\text{CH}_{2\beta}$), 7.34 (1H, dd, $J=8.4$, 1.8 Hz, Ar), 7.39–7.46 (2H, m, Ar), 7.65 (1H, d, $J=1.2$ Hz, Ar), 7.75–7.80 (3H, m, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.7, 27.6, 48.3, 52.0, 124.1, 125.6, 126.0, 126.8 (2C), 127.7, 127.9, 128.0, 132.5, 133.7, 140.4, 144.0, 167.7; ν_{max} (thin film)/ cm^{-1} 2962m, 1721s, 1627m, 1436m, 1256s, 1151s, 947m; m/z (EI) 254 (M^+ , 70%), 194 (30), 165 (100). [Found (HRMS, EI) M^+ 254.1298. $\text{C}_{17}\text{H}_{18}\text{O}_2$ requires M 254.1307].

4.7.4. Methyl-2-methylene-3-(4-methoxyphenyl)-pentanoate

10d. Yield 91%. (Found: C, 71.81; H, 7.73%. Calcd for $\text{C}_{14}\text{H}_{18}\text{O}_3$: C, 71.77; H, 7.74%); δ_{H} (400 MHz, CDCl_3) 0.85 (3H, t, $J=7.3$ Hz, CH_2Me), 1.67–1.92 (2H, m, CH_2Me), 3.67 (3H, s, CO_2Me), 3.78 (3H, s, ArOMe), 5.62 (1H, s, $=\text{CH}_{2\alpha}$), 6.25 (1H, s, $=\text{CH}_{2\beta}$), 6.80–6.84 (2H, m, Ar), 7.10–7.13 (2H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.6, 27.7, 47.4, 52.0, 55.4, 113.8 (2C), 123.6, 129.2 (2C), 134.9, 144.3; ν_{max} (thin film)/ cm^{-1} 2960m, 1722s, 1627m, 1610m, 1253s, 1038m, 946m; m/z (EI) 234 (M^+ , 73%), 205 (100). [Found (HRMS, EI) M^+ 234.1261. $\text{C}_{14}\text{H}_{18}\text{O}_3$ requires M 234.1256].

4.7.5. Methyl-2-methylene-3-(4-tolyl)-pentanoate

10e. Yield 97%. (Found: C, 76.99; H, 8.27%. Calcd for $\text{C}_{14}\text{H}_{18}\text{O}_2$: C, 77.03; H, 8.31%); δ_{H} (400 MHz, CDCl_3) 0.85 (3H, t, $J=7.3$ Hz, CH_2Me), 1.69–1.92 (2H, m, CH_2Me), 2.30 (3H, s, ArMe), 3.66 (3H, s, OMe), 3.70 (1H, dd, $J=8.9$, 6.4 Hz, CH), 5.63 (1H, s, $=\text{CH}_{2\alpha}$), 6.27 (1H, s, $=\text{CH}_{2\beta}$), 7.09 (4H, s, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.6, 21.2, 27.7, 47.8, 52.0, 123.8, 128.2, 129.2, 136.0, 139.9, 144.2, 167.8; ν_{max} (thin film)/ cm^{-1} 2962m, 1721s, 1627m, 1437m, 1151s, 945m; m/z (EI) 218 (M^+ , 61%), 186 (38), 158 (40), 129 (100). [Found (HRMS, EI) M^+ 218.1305. $\text{C}_{14}\text{H}_{18}\text{O}_2$ requires M 218.1307].

4.7.6. Methyl-2-methylene-3-(2-bromophenyl)-pentanoate

10f. Yield 90%; δ_{H} (400 MHz, CDCl_3) 0.92 (3H, t, $J=7.3$ Hz, CH_2Me), 1.71–1.90 (2H, m, CH_2Me), 3.68 (3H, s, OMe), 4.29 (1H, br dd, $J=7.4$ Hz, and unresolved, CH), 5.59 (1H, s, $=\text{CH}_{2\alpha}$), 6.37 (1H, s, $=\text{CH}_{2\beta}$), 7.05, (1H, ddd, $J=8.0$, 7.2, 1.8 Hz, Ar), 7.15 (1H, dd, $J=7.8$, 1.8 Hz, Ar), 7.24 (1H, ddd, $J=7.8$, 7.2, 1.3 Hz, Ar), 7.46 (1H, dd, $J=8.0$, 1.3 Hz, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.3, 27.6, 46.6, 52.1, 125.1, 126.1, 127.5, 128.0, 128.4, 133.2, 142.1, 142.9,

167.5; ν_{max} (thin film)/ cm^{-1} 2964s, 2934m, 2875m, 1724s, 1628m, 1468s, 1252s, 1153s, 948m; m/z (CI) 283 ($[\text{M}+\text{H}]^+$, ^{79}Br , 100%). [Found (HRMS, CI) $[\text{M}+\text{H}]^+$ 283.0317. $\text{C}_{13}\text{H}_{15}\text{O}_2\text{Br}$ requires $\text{M}+\text{H}$ 283.0334].

4.7.7. Methyl-2-methylene-3-(4-chlorophenyl)-pentanoate

10g. Yield 95%. (Found: C, 65.43; H, 6.30%. Calcd for $\text{C}_{13}\text{H}_{15}\text{O}_2\text{Cl}$: C, 65.41; H, 6.33%); δ_{H} (400 MHz, CDCl_3) 0.85 (3H, t, $J=7.3$ Hz, CH_2Me), 1.67–1.93 (2H, m, CH_2Me), 3.67 (3H, s, OMe), 3.70 (1H, dd, $J=9.0$, 6.9 Hz, CH), 5.65 (1H, s, $=\text{CH}_{2\alpha}$), 6.30 (1H, s, $=\text{CH}_{2\beta}$), 7.12–7.15 (2H, m, Ar), 7.23–7.26 (2H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.5, 27.5, 47.6, 52.1, 124.2, 128.6, 129.6, 132.2, 141.4, 143.6, 167.5; ν_{max} (thin film)/ cm^{-1} 2964m, 1721s, 1627m, 1437m, 1152s, 948m; m/z (EI) 238 (M^+ , ^{35}Cl , 100%). [Found (HRMS, EI) M^+ 238.0765. $\text{C}_{13}\text{H}_{15}\text{O}_2\text{Cl}$ requires M 238.0761].

4.7.8. Methyl-2-methylene-3-(4-fluorophenyl)-pentanoate

10h. Yield 94%. (Found: C, 70.23; H, 6.75%. Calcd for $\text{C}_{13}\text{H}_{15}\text{O}_2\text{F}$: C, 70.25; H, 6.80%); δ_{H} (400 MHz, CDCl_3) 0.85 (3H, t, $J=7.3$ Hz, CH_2Me), 1.67–1.93 (2H, m, CH_2Me), 3.67 (3H, s, OMe), 3.71 (1H, dd, $J=9.0$, 6.3 Hz, CH), 5.64 (1H, s, $=\text{CH}_{2\alpha}$), 6.29 (1H, s, $=\text{CH}_{2\beta}$), 6.93–6.99 (2H, m, Ar), 7.13–7.18 (2H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) 12.5, 27.7, 47.5, 52.0, 115.2 (d, $^2J_{\text{CF}}=21.2$ Hz), 124.0, 129.7 (d, $^3J_{\text{CF}}=7.8$ Hz), 138.5 (d, $^4J_{\text{CF}}=3.1$ Hz), 143.9, 161.6 (d, $^1J_{\text{CF}}=244.0$ Hz), 167.6; ν_{max} (thin film)/ cm^{-1} 2964m, 1723s, 1628m, 1603m, 1508s, 1151s, 948m; m/z (EI) 222 (M^+ , 93%), 190 (55), 162 (32), 147 (23), 133 (100). [Found (HRMS, EI) M^+ 222.1060. $\text{C}_{13}\text{H}_{15}\text{O}_2\text{F}$ requires M 222.1056].

4.7.9. Methyl-2-methylene-3-(4-trifluoromethylphenyl)-pentanoate

10i. Yield 85%; δ_{H} (400 MHz, CDCl_3) 0.87 (3H, t, $J=7.3$ Hz, CH_2Me), 1.73–1.95 (2H, m, CH_2Me), 3.67 (3H, s, OMe), 3.79 (1H, dd, $J=8.4$, 6.8 Hz, CH), 5.71 (1H, s, $=\text{CH}_{2\alpha}$), 6.35 (1H, s, $=\text{CH}_{2\beta}$), 7.32 (2H, d, $J=8.2$ Hz, Ar), 7.53 (2H, d, $J=8.2$ Hz, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.4, 27.4, 48.1, 52.1, 124.5 (q, $^1J_{\text{CF}}=272$ Hz), 124.6, 125.4 (q, $^3J_{\text{CF}}=3.8$ Hz), 128.6, 128.8 (q, $^2J_{\text{CF}}=32.2$ Hz), 143.2, 147.2, 167.3; ν_{max} (thin film)/ cm^{-1} 2937m, 2878m, 1725s, 1618s, 1439s, 1325s, 1255s, 1163s, 1123s, 1068s, 1019m, 834m; m/z (EI) 272 (M^+ , 100%). [Found (HRMS, EI) M^+ 272.1013. $\text{C}_{14}\text{H}_{15}\text{O}_2\text{F}_3$ requires M 272.1024].

4.7.10. Methyl-2-methylene-3-(4-nitrophenyl)-pentanoate

10j. Yield 97%; δ_{H} (400 MHz, CDCl_3) 0.87 (3H, t, $J=7.3$ Hz, CH_2Me), 1.75–1.98 (2H, m, CH_2Me), 3.67 (3H, s, OMe), 3.84 (1H, br dd, $J=7.6$ Hz, and unresolved, CH), 5.76 (1H, s, $=\text{CH}_{2\alpha}$), 6.40 (1H, s, $=\text{CH}_{2\beta}$), 7.36–7.39 (2H, m, Ar), 8.13–8.18 (2H, m, Ar); δ_{C} (100.6 MHz, CDCl_3) 12.4, 27.3, 48.2, 52.2, 123.8, 125.2, 129.1 (2C), 142.6, 150.9, 167.1; ν_{max} (thin film)/ cm^{-1} 2964m, 2876m, 1720s, 1628m, 1519s, 1347s, 1256s, 1153s, 953m, 842m, 714m; m/z (EI) 249 (M^+ , 70%), 232 (100). [Found (HRMS, EI) M^+ 249.0998. $\text{C}_{13}\text{H}_{15}\text{NO}_4$ requires M 249.1001].

4.7.11. tert-Butyl-2-methylene-3-phenyl-pentanoate

10k. Yield 83%; colourless oil; δ_{H} (400 MHz, CDCl_3) 0.86 (3H, t, $J=7.3$ Hz, CH_2Me), 1.35 (9H, s, CMe_3), 1.67–1.93 (2H, m, CH_2Me), 3.65 (1H, dd, $J=8.9$, 6.1 Hz, CH), 5.53 (1H, s,

$=CH_{2\alpha}$), 6.20 (1H, s, $=CH_{2\beta}$), 7.15–7.29 (5H, m, Ar); δ_C (100.6 MHz, $CDCl_3$): 12.6, 27.6, 28.1, 48.3, 80.8, 122.8, 126.3, 128.3 (2C), 128.4 (2C), 143.2, 145.8, 166.7; ν_{max} (thin film)/ cm^{-1} 2932s, 2875s, 1713s, 1628s, 1453s, 1392s, 1367s, 1254s, 1148s, 943m; m/z (EI) 246 (M^+ , 11%), 190 (100). [Found (HRMS, EI) M^+ 246.1632. $C_{16}H_{22}O_2$ requires M 246.1620].

4.7.12. Methyl-2-methylene-3-phenyl-butanoate 11a.

Yield 85%. (Found: C, 75.75; H, 7.37%. Calcd for $C_{12}H_{14}O_2$: C, 75.76; H, 7.42%); δ_H (400 MHz, $CDCl_3$) 1.42 (3H, d, $J=7.2$ Hz, $CHMe$), 3.67 (3H, s, OMe), 4.03 (1H, q, $J=7.2$ Hz, $CHMe$), 5.61 (1H, s, $=CH_{2\alpha}$), 6.28 (1H, s, $=CH_{2\beta}$), 7.18–7.30 (5H, m, Ar); δ_C (100.6 MHz, $CDCl_3$) 21.0, 40.7, 52.0, 124.0, 126.5, 127.6, 128.6, 144.5, 145.1, 167.6; ν_{max} (thin film)/ cm^{-1} 2969m, 1721s, 1627m, 1437m, 1149s, 947m; m/z (EI) 190 (M^+ , 92%), 158 (42), 130 (100). [Found (HRMS, EI) M^+ 190.0997. $C_{12}H_{14}O_2$ requires M 190.0994].

4.7.13. Methyl-2-methylene-3-phenyl-heptanoate 11b.

Yield 98%; δ_H (400 MHz, $CDCl_3$) 0.85 (3H, t, $J=7.2$ Hz, CH_2Me), 1.11–1.32 (4H, m, CH_2CH_2Me), 1.69–1.88 (2H, m, $CHCH_2$), 3.67 (3H, s, OMe), 3.83 (1H, dd, $J=7.2$, 6.9 Hz), 5.64 (1H, s, $=CH_{2\alpha}$), 6.28 (1H, s, $=CH_{2\beta}$), 7.16–7.36 (5H, m, Ar); δ_C (67.8 MHz, $CDCl_3$) 14.2, 22.8, 30.1, 34.4, 46.3, 52.0, 124.0, 126.5, 128.3, 128.5, 143.1, 144.2, 167.7; ν_{max} (thin film)/ cm^{-1} 2954s, 2931s, 1723s, 1627m, 1437m, 1149s, 945m; m/z (EI) 232 (M^+ , 53%), 201 (62), 175 (31), 130 (47), 115 (100). [Found (HRMS, EI) M^+ 232.1471. $C_{15}H_{20}O_2$ requires M 232.1463].

4.7.14. Methyl-2-methylene-3-phenyl-hexanoate 11c.

Yield 70%; δ_H (400 MHz, $CDCl_3$) 0.89 (3H, t, $J=7.3$ Hz, CH_2Me), 1.17–1.30 (2H, m, CH_2Me), 1.69–1.85 (2H, m,

$CHCH_2$), 3.67 (3H, s, OMe), 3.85 (1H, m, CH), 5.65 (1H, s, $=CH_{2\alpha}$), 6.28 (1H, s, $=CH_{2\beta}$), 7.18–7.36 (5H, m, Ar); δ_C (100.6 MHz, $CDCl_3$) 14.2, 21.0, 36.9, 46.1, 52.0, 123.9, 126.5, 128.3, 128.4, 143.1, 144.2, 167.7; ν_{max} (thin film)/ cm^{-1} 2955s, 2871m, 1718s, 1627m, 1437m, 1244s, 1149s, 944m; m/z (EI) 218 (M^+ , 51%), 187 (55), 158 (19), 129 (34), 115 (100). [Found (HRMS, EI) M^+ 218.1308. $C_{14}H_{18}O_2$ requires M 218.1307].

4.7.15. Ethyl-4-(dimethylmethoxysilyl)-2-methylene-3-phenyl-butanoate 11d.

The reagent $[Cu(MeCN)_4]BF_4$ (8 mg, 0.015 mmol) was dissolved in dry THF (1.0 ml) at $-30^\circ C$ under an argon atmosphere and $Zn(CH_2SiMe_2-OMe)_2$ (2.0 ml of 0.25 M THF/pentane, 0.50 mmol) and allyl bromide **3a** (135 mg, 0.50 mmol, dissolved in dry degassed THF (1.0 ml) were added at the same time via syringe pump over 20 min. The reaction was allowed to warm to $-5^\circ C$ slowly over about 40 min, then quenched with water. Standard workup with ether and purification by flash chromatography (diethylether/light petroleum, 1:1) gave the desired compound. Yield 93%, oil; δ_H ($CDCl_3$, 400 MHz) -0.12 , -0.05 (6H, $2 \times SiMe$), 1.20 (3H, t, $J=6.9$ Hz, CH_2Me), 1.24 (2H, m, CH_2-Si), 3.31 (3H, s, $SiOMe$), 4.10 (3H, m, CH_2Me overlapped by CH), 5.70 (1H, s, $=CH_{2\alpha}$), 6.25 (1H, $=CH_{2\beta}$), 7.14–7.28 (5H, m, Ar); δ_C ($CDCl_3$, 67.8 MHz) -2.3 , -2.0 , 14.0, 22.2, 41.4, 60.5, 123.0, 126.3, 128.1 (2C), 143.9, 146.2, 167.2; ν_{max} (thin film)/ cm^{-1} 2930s, 2874s, 1697s, 1628s, 1466m, 1430m, 1361m, 1254m, 1177m; m/z (EI) 261 ($[M-OMe]^+$, 18%) [Found (HRMS, EI): $[M-OMe]^+$, 261.1318 $C_{16}H_{24}O_3Si$ requires $[M-OMe]$ 261.1311].

The required organozinc was prepared from sodium distilled $MeOSiMe_3$ (2.76 ml, 20 mmol) in THF (7.5 ml) and Bu^tLi (12.1 ml of 1.65 M pentane solution, 20.0 mmol)

Table 4. Separation of enantiomers **5** and **10**

Compound	Conditions	Retention times/min
5a (Ph)	GC: 2,6-Me-3-pe- γ -CD; He carrier gas, 12 psi ^a	11.0 (+) ^b 17.0 (–)
5b (1-Nap)	HPLC: Chiralcel OD; 99:1 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	13.5 (–) 36.1(+)
5c (4-MePh)	HPLC: Chiralcel OD; 99:1 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	11.9 (+) 13.4 (–)
5e (4-NO ₂ Ph)	HPLC: Chiralcel OD; 99:1 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	23.9 (+) 25.7 (–)
10a (Ph)	GC: 2,6-Me-3-pe- δ -CD; He carrier gas, 12 psi	21.8 22.8
10b (1-Nap)	HPLC: Chiralcel OD-H; 90:10 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	9.5 17.4
10d (4-MeOPh)	HPLC: Chiralcel OD; 99.5:0.5 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	15.7 20.4
10e (4-MePh)	HPLC: Chiralcel OD; 99.8:0.2 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	14.1 24.8
10f (2-BrPh)	HPLC: Chiralcel OD; 98:2 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	11.1 23.0
10h (4-FPh)	HPLC: Chiralcel OD; 99.9:0.1 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	17.0 23.0
10i (4-CF ₃ Ph)	GC: Cyclodextrin-B; He carrier gas, 12 psi	51.8 52.9
10j (4-NO ₂ Ph)	HPLC: Chiralcel OD; 99.5:0.5 hexane/isopropanol; flow rate of 0.5 ml min ^{–1}	22.6 24.7
10k (Ph)	GC: 2,6-Me-3-pe- γ -CD; He carrier gas, 12 psi	31.9 33.6

^a 2,6-Me-3-pe- δ -CD is a 25 m octakis(2,6-di-*O*-methyl-3-*O*-pentyl)- γ -cyclodextrin column.¹⁷

^b Sign of optical rotation taken from preliminary studies.⁵

lithiated at -78°C followed by warming to ambient and further reaction (2 h).¹⁰ A solution of fused ZnCl_2 (1.36 g, 10.0 mmol) in THF (16 ml) was added slowly to the $\text{LiCH}_2\text{SiMe}_2\text{OMe}$ and the mixture was Schlenk filtered after overnight stirring to afford a 0.25 M solution (Gilman titration).

4.7.16. Ethyl 2-methylene-3-phenyl-4-trimethylsilylbutyrate 11e. Prepared in an identical manner to compound **8a**. The required $\text{Zn}(\text{CH}_2\text{TMS})_2$ was prepared by literature routes.¹⁶ Yield 25%; δ_{H} (400 MHz, CDCl_3) -0.15 (9H, s, CH_2SiMe_3), 1.13 – 1.14 (2H, m, CH_2SiMe_3), 1.21 (3H, t, $J=7.0$ Hz, OCH_2Me), 4.02 (1H, dd, $J=8.1, 7.8$ Hz, CH), 4.06 – 4.18 (2H, m, OCH_2Me), 5.69 (1H, s, $=\text{CH}_{2\alpha}$), 6.23 (1H, s, $=\text{CH}_{2\beta}$), 7.14 – 7.28 (5H, m, Ar); δ_{C} (67.8 MHz, CDCl_3) $-1.24, 14.0, 22.8, 42.2, 60.5, 122.8, 126.2, 128.1$ (2C), $144.1, 146.5, 167.0$; ν_{max} (thin film)/ cm^{-1} 2952m, 1718s, 1261m, 1248m, 1173m, 1150m; m/z (CI) 276 (M^+ , 29%), 263 (22), 262 (31), 261 (100). [Found (HRMS, CI): M^+ 276.1556. $\text{C}_{16}\text{H}_{24}\text{O}_2\text{Si}$ requires M 276.1546] (Table 4).

Acknowledgements

The European Union is thanked for its support through COST-D24 and the EPSRC for grant GR/S56054. We are grateful to the Grundy Educational Trust and the University of Nottingham for the support of a studentship (PJG). The contribution of the EPSRC Mass Spectrometry Service is also gratefully acknowledged.

References and notes

- Recent overviews of Diels–Alder chemistry: (a) Hayashi, Y. In *Cycloaddition Reactions in Organic Synthesis*; Kobayashi, S., Jorgensen, K. A., Eds.; Wiley-VCH: Germany, 2002; pp 5–55. (b) Nicolaou, K. C.; Snyder, S. A.; Montagnon, T.; Vassilikogiannakis, G. *Angew. Chem., Int. Ed.* **2002**, *41*, 1668–1698. (c) Nicolaou, K. C.; Montagnon, T.; Snyder, S. A. *Chem. Commun.* **2003**, 551–564.
- Perlmutter, P. *Conjugate Addition Reactions in Organic Synthesis*; Pergamon: Oxford, 1992.
- Rare examples: (a) Anirban, A. N. P. *Tetrahedron* **2003**, *59*, 2991–2998. (b) Ameer, F.; Drewes, S. E.; Hoole, R.; Kaye, P. T.; Pitchford, A. T. *J. Chem. Soc. Perkin Trans. 1* **1985**, 2713–2718. Reaction with aldehydes is more common, recent examples: (c) Paquette, L. A. *Synthesis* **2003**, 765–774. (d) Lee, A. S.-Y.; Chan, Y.-T.; Wang, S.-H.; Chu, S.-H. *Tetrahedron Lett.* **2002**, *43*, 8489–8492.
- Rare example: (a) Doucet, H.; Brown, J. M. *Bull. Chem. Soc. Fr.* **1997**, *134*, 995–999. Reaction with carbonyl/imine electrophiles is more common, recent example: (b) Sergeeva, N. N.; Golubev, A. S.; Hennig, L.; Findeisen, M.; Pactzold, E.; Oehme, G.; Burger, K. *J. Fluor. Chem.* **2001**, *111*, 41–44.
- Börner, C.; Gimeno, J.; Gladiali, S.; Goldsmith, P. J.; Ramazzotti, D.; Woodward, S. *Chem. Commun.* **2000**, 2433–2434.
- Xu, L.-H.; Kündig, E. P. *Helv. Chim. Acta* **1994**, *77*, 1480–1484.
- For reviews covering promotion of Baylis–Hillman procedures see: (a) Basavaiah, D.; Rao, D.; Satyanarayana, T. *Chem. Rev.* **2003**, *103*, 811–891. (b) Basavaiah, D.; Rao, P. D.; Hyma, R. S. *Tetrahedron* **1996**, *52*, 8001–8062. (c) Aggarwal, V. K.; Emme, V. K.; Fulford, I. J. *Org. Chem.* **2003**, *68*, 692–700.
- Buchholz, R.; Hoffmann, H. M. R. *Helv. Chim. Acta* **1991**, *74*, 1213–1220.
- Feringa, B. L. *Acc. Chem. Res.* **2000**, *33*, 346–353.
- Bates, T. F.; Dandekar, S. A.; Longlet, J. J.; Wood, K. A.; Thomas, R. D. *J. Organomet. Chem.* **2000**, *595*, 87–92.
- Most relevant review: (a) Drews, S. E.; Roos, G. H. P. *Tetrahedron* **1988**, *44*, 4653–4670. For the specific synthesis and spectroscopic data see: (b) Fort, Y.; Berthe, M. C.; Caubere, P. *Tetrahedron* **1992**, *48*, 6371–6384 for **2a**. (c) Lawrence, N. J.; Crump, J. P.; McGown, A. T.; Hadfield, J. A. *Tetrahedron Lett.* **2001**, *42*, 3939–3942 for **2c** and **4c**. (d) Ciclosi, M.; Fava, C.; Galeazzi, R.; Orena, M.; Sepulveda-Arques, J. *Tetrahedron Lett.* **2002**, *43*, 2199–2202 for **2d**. (e) Basavaiah, D.; Krishnamacharyulu, M.; Rao, J. *Synth. Commun.* **2000**, *30*, 2061–2069. (f) Lee, K. Y.; Kim, J. M.; Kim, J. N. *Tetrahedron* **2003**, *59*, 385–390 for **2g**. (g) Amri, H.; Beltaief, I.; Besbes, R.; Amri, H.; Villéras, M.; Villéras, J. *Synthesis* **1998**, 1765–1769 for **2h** and precursor to **7**. (h) Coelho, F.; Almeida, W. P.; Veronese, D.; Mateus, C. R.; Lopes, E. C.; Rossi, R. C.; Silveria, G. P. C.; Pavam, C. H. *Tetrahedron* **2002**, *58*, 7437–7448 for **8a**, **8d**, **8f–g**, **8j**. (i) Lawrence, R.; Perlmutter, P. *Chem. Lett.* **1992**, 305–308 for **8b**. (j) Nascimento, M. G.; Zanutto, S. P.; Melegari, S. P.; Fernandes, L.; Sa, M. M. *Tetrahedron: Asymmetry* **2003**, *14*, 3111–3116 for **8c**. (k) Sammelson, R. E.; Gurusinge, C. D.; Kurth, J. M.; Olmstead, M. M.; Kurth, M. J. *J. Org. Chem.* **2002**, *67*, 876–882 for **8e**. (l) Yu, C.; Liu, B.; Hu, L. *J. Org. Chem.* **2001**, *66*, 5413–5418. (m) Cai, J.; Zhou, Z.; Zhao, G.; Tang, C. *Org. Lett.* **2002**, *4*, 4723–4726 for **8i**. (n) Hoffmann, H. M. R.; Gassner, A.; Eggert, U. *Chem. Ber.* **1991**, *124*, 2475–2480 for **8k**.
- (a) Lambert, F.; Kirschleger, B.; Villieras, J. *J. Organomet. Chem.* **1991**, *406*, 71–86 for **3a** and **3d**. (b) Du, Y.; Lu, X.; Zhang, C. *Angew. Chem., Int. Ed.* **2003**, *42*, 1035–1037 for **3c**. (c) Ameer, F.; Drewes, S. E.; Emslie, N. D.; Kaye, P. T.; Mann, R. L. *J. Chem. Soc., Perkin Trans. 1* **1983**, 2293–2295 for **3a** and **3e**.
- (a) Chavan, S. P.; Ethiraj, K. S.; Kamat, S. K. *Tetrahedron Lett.* **1997**, *38*, 7415–7416 for **4a–b**, **4d–e**, and **4g**. (b) Yuasa, Y.; Yuasa, Y.; Tsuruta, H. *Aust. J. Chem.* **1998**, *51*, 511–514 for **9a**. (c) Yadav, V. K.; Babu, K. G. *Tetrahedron* **2003**, *59*, 9111–9116 for **9d**.
- Bennett, S. M. W.; Brown, S. M.; Cunningham, A.; Dennis, M. R.; Muxworthy, J. P.; Oakley, M. A.; Woodward, S. *Tetrahedron* **2000**, *56*, 2847–2855.
- Fournie-Zaluski, M.-C.; Coric, P.; Turcaud, S.; Rousselet, N.; Gonzalez, W.; Barbe, B.; Pham, I.; Jullian, N.; Michel, J.-B.; Roques, B. P. *J. Med. Chem.* **1994**, *37*, 1070–1083.
- Westerhausen, M.; Rademacher, B.; Poll, W. *J. Organomet. Chem.* **1991**, *421*, 175–188.
- König, W. A.; Gehrcke, B.; Icheln, D.; Evers, P.; Doennecke, J.; Wang, W. *J. High Res. Chromatogr.* **1992**, *15*, 367–372.